

*** Unit 3 Lecture***

✓ ***This script will be a part of HW #2 Functions and Modules***

Slide 1: Introduction to Functions

```
print("Functions allow us to reuse code and make our programs more modular.")
```

Functions allow us to reuse code and make our programs more modular.

Slide 2-3: Defining and Calling a Simple Function

```
def greet():  
    print("Hello! Welcome to Python Functions.")  
  
# Run the function  
greet()
```

Hello! Welcome to Python Functions.

Challenge 1

```
# Vikram Nadathur May 28th 2026  
# Challenge 1: Define a function that prints "Python is fun!"  
# Complete the code below  
def python_fun():  
    print("Python is fun!")  
python_fun()
```

Python is fun!

Slides 6-7: Function with Parameters

```
def personalized_greeting(name):  
    print(f"Hello, {name}! Welcome to Python Functions.")  
  
# Run the function  
personalized_greeting("Alice")
```

Hello, Alice! Welcome to Python Functions.

Challenge 2: Define a function that takes two arguments and prints their sum

```
# Vikram Nadathur May 28th 2026  
# Complete the code below  
def print_sum(a, b):  
    return(a+b)  
print_sum(3,5)
```

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Slide 7: Returning Values from Functions

```
def add(a, b):  
    return a + b  
  
# Run the function and print the result  
result = add(3, 5)  
print(f"The sum is: {result}")
```

The sum is: 8

Challenge 3: Define a function that returns the product of two numbers

```
# Vikram Nadathur May 28th 2026  
# Complete the code below  
def multiply(a, b):  
    return(a*b)  
multiply(3,5)
```

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Slide 8-9: Local and Global Variables

```
# Example 4: Local Variable  
def local_variable_example():  
    local_var = 10  
    print(f"Local variable: {local_var}")  
  
# Run the function  
local_variable_example()
```

Local variable: 10

Challenge 4: Define a function that declares a local variable and prints it

```
# Vikram Nadathur May 28th 2026  
# Complete the code below  
def print_local():  
    local_var = 10  
    print(f"Local variable: {local_var}")  
print_local()
```

Local variable: 10

Slide 10: Global Variable

```
global_var = 20  
  
def global_variable_example():  
    print(f"Global variable: {global_var}")
```

```
# Run the function
global_variable_example()
```

Global variable: 20

Challenge 5: Modify the global variable within a function

```
# Vikram Nadathur May 28th 2026
# Complete the code below
def modify_global():
    global global_var
    global_var = 30
modify_global()
print(global_var)
```

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Slide 11-13: Passing Arguments to Functions

```
# Example 6: Passing Multiple Arguments
def print_full_name(first_name, last_name):
    print(f"Full name: {first_name} {last_name}")

# Run the function
print_full_name("John", "Doe")
```

Full name: John Doe

Challenge 6: Define a function that takes three arguments and prints them

```
# Vikram Nadathur May 28th 2026
# Complete the code below
def print_three(a, b, c):
    print(a, b, c)

# Example call
print_three(1, 2, 3)
```

1 2 3

Slide 14-15: Keyword Arguments

```
def describe_pet(animal_type, pet_name):
    print(f"\nI have a {animal_type}.")
    print(f"My {animal_type}'s name is {pet_name}.")

# Run the function using keyword arguments
describe_pet(animal_type='hamster', pet_name='Harry')
```

I have a hamster.
My hamster's name is Harry.

Challenge 7: Define a function with keyword arguments and call it

```
# Vikram Nadathur May 28th 2026
```

```
# Vikram Nadathur May 28th 2026
# Complete the code below
def show_vehicle(type, brand):
    print(f"My vehicle is a {brand} {type}.")

# Example calls using keyword arguments
show_vehicle(type='car', brand='Toyota')
show_vehicle(brand='Tesla', type='truck')
```

My vehicle is a Toyota car.
My vehicle is a Tesla truck.

Slide 10-12: Default Argument Values

```
# Example 8: Default Argument
def print_greeting(name, greeting="Hello"):
    print(f"{greeting}, {name}!")

# Run the function with and without the default argument
print_greeting("Alice")
print_greeting("Bob", "Hi")
```

Hello, Alice!
Hi, Bob!

Challenge 8: Define a function with a default argument and call it

```
# Vikram Nadathur May 28th 2026
# Complete the code below

def introduce(name, age=25):
    print(f"My name is {name} and I am {age} years old.")

# Example calls
introduce("Alice")
introduce("Bob", 30)
```

My name is Alice and I am 25 years old.
My name is Bob and I am 30 years old.

Slide 13: Value-Returning Functions

```
# Example 9: Value-Returning Function
def square(number):
    return number * number

# Run the function and print the result
result = square(4)
print(f"The square of 4 is: {result}")
```

The square of 4 is: 16

Challenge 9: Define a function that returns the cube of a number

```
# Vikram Nadathur May 28th 2026
# Complete the code below
```

```
def cube(number):  
    return number * number * number  
  
# Example call  
print(cube(3))
```

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Slides 14-16: The random and math Modules

```
import random  
# Generate a random float in the range [0.0, 1.0)  
number = random.random()  
print("Random float between 0.0 and 1.0:", number)
```

Random float between 0.0 and 1.0: 0.30036210324236257

```
import random  
# Generate a random integer between 1 and 100  
number = random.randint(1, 100)  
print("Random number between 1 and 100:", number)
```

Random number between 1 and 100: 37

```
import math  
  
# Example 10: Using the math Module  
def calculate_circle_area(radius):  
    return math.pi * radius * radius  
  
# Run the function and print the result  
area = calculate_circle_area(5)  
print(f"The area of the circle is: {area:.2f}")
```

The area of the circle is: 78.54

Challenge 10: Define a function that uses the math module to calculate the square root of a number

```
# Vikram Nadathur May 28th 2026  
# Complete the code below  
  
import math  
  
def calculate_sqrt(number):  
    return math.sqrt(number)  
  
# Example call  
print(calculate_sqrt(16))
```

4.0

Slide 20-21: Storing Functions in Modules

```
# Vikram Nadathur May 28th 2026  
# Complete the code below by creating with your functions
```

```
%%writefile custom_module.py
def function_one():
    print("Function One")

def function_two():
    print("Function Two")
```

Writing custom_module.py

uncommend 2 lines below

```
%%writefile my_module.py
def my_function():
    print("Hello from my_module!")
```

Writing my_module.py

```
from my_module import my_function
my_function()
```

Hello from my_module!

Challenge 11: Create your own module with at least two functions and import it here

```
# Vikram Nadathur May 28th 2026
# Complete the code below be creating with your functions
```

```
%%writefile custom_module.py
def function_one():
    print("Function One")

def function_two():
    print("Function Two")
```

Overwriting custom_module.py

```
from custom_module import function_one, function_two
```

```
function_one()
function_two()
```

Function One
Function Two

```
# Example 14: Generating Random Numbers
import random
```

```
def roll_die():
    return random.randint(1, 6)
```

```
# Run the function and print the result
die_roll = roll_die()
print(f"Rolled a die: {die_roll}")
```

Rolled a die: 1

Challenge 13: Define a function that generates a random float between two numbers

```
# Vikram Nadathur May 28th 2026
# Complete the code below

import random

def random_float(min_value, max_value):
    return random.uniform(min_value, max_value)

# Example call
print(random_float(1.5, 4.2))
```

3.0307681949089726

```
# Example 15: Returning Multiple Values
def min_max(numbers):
    return min(numbers), max(numbers)

# Run the function and print the result
min_val, max_val = min_max([1, 2, 3, 4, 5])
print(f"Min: {min_val}, Max: {max_val}")
```

Min: 1, Max: 5

Challenge 14: Define a function that returns the sum, difference, and product of two numbers

```
# Vikram Nadathur May 28th 2026
# Complete the code below

def calculate(a, b):
    total = a + b
    difference = a - b
    product = a * b
    return total, difference, product

# Example call
s, d, p = calculate(5, 3)
print("Sum:", s, "Difference:", d, "Product:", p)
```

Sum: 8 Difference: 2 Product: 15

Returning None from a Function

```
def divide(num1, num2):
    if num2 == 0:
        return None
    else:
        return num1 / num2

# Run the function and handle the result
result = divide(10, 0)
if result is None:
    print("Cannot divide by zero!")
else:
    print(f"Result: {result}")
```

Cannot divide by zero!

Challenge 15: Define a function that returns None if a number is negative

```
# Vikram Nadathur May 28th 2026
# Complete the code below

def check_positive(number):
    if number < 0:
        return None
    else:
        return number

# Example calls
print(check_positive(10))    # 10
print(check_positive(-5))   # None
```

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None

Conditional Execution of the main Function

```
def main():
    print("This is the main function.")

def secondary_function():
    print("This is a secondary function.")

if __name__ == '__main__':
    main()
```

This is the main function.

Challenge 16: Define two functions and conditionally execute one based on a variable value

```
# Vikram Nadathur May 28th 2026
# Complete the code below

def function_one():
    print("Function one is running.")

def function_two():
    print("Function two is running.")

# Choose which function to run
```



```
run_first = True # change to False to run function_two instead

if run_first:
    function_one()
else:
    function_two()
```

Function one is running.

```
%%writefile nasa.py
# apod.py
import requests
from IPython.display import display, Image

def get_apod(api_key):
    response = requests.get(f'https://api.nasa.gov/planetary/apod?api_key={api_key}')
    if response.status_code == 200:
        data = response.json()
        return data['title'], data['url'], data['explanation']
    else:
        return "Could not fetch APOD data", "", ""

def display_apod(api_key):
    title, url, explanation = get_apod(api_key)
    if title == "Could not fetch APOD data":
        print(title)
    else:
        print(f"Title: {title}\nURL: {url}\nExplanation: {explanation}")
        display(Image(url=url, embed=True))
```

Writing nasa.py

```
# Import the apod module
import nasa

# Use the DEMO_KEY provided by NASA
api_key = "DEMO_KEY"

# Call the function to display APOD data and image
nasa.display_apod(api_key)
```

Title: NGC 1514: The Crystal Ball Nebula

URL: <https://apod.nasa.gov/apod/image/2605/noirlab2613a.jpg>

Explanation: What do you see in this crystal ball? The featured image shows NGC 1514, known



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